

MLIM — Memory Learning Integrity Module

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Independent Methodological Authority

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Architecture Ecosystem: Structured Reality Standards™

Architecture Family: Memory Governance Intelligence Architecture (MGIA™)

Operational Layer: Memory Evolution Governance Layer

Governed Space: Memory Learning Integrity

Category: AI & Interpretation

Subcategory: Memory Learning Governance

Type: Memory Evolution Integrity Module

Parent Standard: Memory Evolution Integrity Standard (MEIS)

Version: 1.0

Status: Canonical · Open Module

Origin Date: 12 June 2026

Compatibility: OOF Methodology OS · Memory Evolution Integrity Standard (MEIS) ·

Memory Integrity Standard (MIS) · Memory Adaptation Integrity Module (MAIM) ·

Cognitive Evolution Governance Standard (CEGS) · Agent Cognition Integrity Standard (ACIS) ·

INTEGROS® — Integrity Standard

AI-Readable: Yes

Authority: OOF

Protection: MIP — Methodological Intellectual Property

Canonical Language: English (UCL)

Canonical Definition

Memory Learning Integrity Module (MLIM) defines the structural conditions under which memory acquires, incorporates, preserves, refines, and utilizes new knowledge, experience, observations, patterns, and operational understanding in a traceable, reconstructable, accountable, and operationally valid manner.

MLIM governs memory learning.

The module ensures that learning remains understandable, reviewable, and connected to identifiable sources rather than becoming opaque, unverifiable, or operationally unreliable.

A system satisfies MLIM only if:

- learning remains identifiable
- learning sources remain traceable
- learning outcomes remain assessable
- learning degradation remains detectable
- learning governance remains possible
- memory learning remains operationally valid

A memory environment that cannot explain what was learned, how it was learned,

or why it was learned does not satisfy MLIM.

Module Operational Role

MLIM defines the memory-learning governance layer of MEIS.

Its role is to preserve integrity of learning throughout memory evolution.

Module Operational Space

MLIM governs:

- memory learning
- knowledge acquisition
- experience integration
- learning traceability
- learning accountability
- memory refinement
- learning governance
- operational understanding

The module applies wherever memory evolves through learning.

Module Function

The module applies wherever systems must preserve:

- trustworthy learning
- explainable knowledge acquisition
- governance-valid adaptation
- reconstructable learning history
- accountable memory evolution
- reliable intelligence development

Its function is to ensure that learning remains connected to evidence, experience, and operational reality.

Minimum Implementation Framework

1. Define the Memory Learning Object

The organization must define which learning environments require governance.

This may include:

- AI learning systems
 - autonomous agents
 - Human-AI environments
 - organizational learning systems
 - collective intelligence systems
 - robotics environments
 - adaptive memory systems
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- knowledge repositories

2. Define Memory Learning Conditions

The system must define the conditions under which learning remains valid.

This includes:

- learning requirements
- evidence requirements
- traceability requirements
- accountability requirements
- validation requirements
- governance-valid learning conditions

3. Define Learning Degradation Detection Logic

The system must define how learning failures are identified.

This may include:

- unverifiable learning
- corrupted learning
- hallucinated learning
- unsupported knowledge acquisition
- learning drift
- evidence-detached learning

4. Define Operational Response or Governance Logic

The system must define governance logic for learning-integrity failures.

Governance response may include:

- learning review
- evidence validation
- governance intervention
- escalation
- rollback procedures
- learning correction
- operational invalidation where required

5. Preserve Traceability & Restrict Invalid Conditions

The system must preserve reconstructable traceability of:

- learning events
- evidence sources
- validation activities
- governance reviews
- intervention procedures
- resulting memory states

A memory-evolution environment must not remain learning-valid if materially significant learning cannot be reconstructed, validated, reviewed, or governed.

Use Case 1 — Autonomous Learning Agent

Scenario

An autonomous agent continuously acquires new operational knowledge through interaction with reality and experience.

Application

MLIM governs learning traceability, evidence linkage, and accountability of acquired knowledge.

Result

The agent gains stronger explainability and reduced exposure to unreliable learning processes.

Use Case 2 — Enterprise Knowledge Development Program

Scenario

An organization continuously incorporates lessons learned, operational experience, and newly acquired expertise into institutional memory.

Application

MLIM governs knowledge acquisition and preservation of learning integrity.

Result

The organization gains stronger learning quality and improved long-term knowledge reliability.

Canonical Closing Statement

Memory Learning Integrity Module (MLIM) defines the structural conditions under which memory acquires, incorporates, preserves, refines, and utilizes new knowledge, experience, observations, patterns, and operational understanding in a traceable, reconstructable, accountable, and operationally valid manner.

Memory evolution depends on learning, but learning without integrity creates unreliable intelligence. Memory learning integrity therefore becomes a foundational integrity condition of trustworthy adaptive memory systems.

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Canonical Source

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